

Problem 4

A crane is constructed from six identical rods, connected to each other and to the wall by hinges, as shown in Figure E. A load of mass m is suspended from the structure.

- (a) What is the axial force in rod AB ?
- (b) Is rod AB in tension or compression?

Solution

Let the mass of the load be m , so the downward force due to gravity is:

$$F_g = mg$$

We solve the problem using static equilibrium conditions — the sum of forces and moments must be zero.

Assume the structure forms a symmetric truss, and the rods create a series of triangles. Each rod carries part of the load through axial forces.

Equilibrium Conditions At joint B , the vertical equilibrium gives:

$$F_{AB} \sin(\theta) = mg$$

The horizontal equilibrium gives:

$$F_{AC} = F_{AB} \cos(\theta)$$

From geometry, if the rod lengths and angles are symmetric, θ can be determined from the geometry of the structure. For simplicity, assume $\theta = 45^\circ$:

Solve for F_{AB} :

$$F_{AB} = \frac{mg}{\sin(45^\circ)} = mg\sqrt{2}$$

Result The axial force in rod AB is:

$$F_{AB} = mg\sqrt{2}$$

Since the force is directed away from the joint to balance the load, rod AB is in tension.

Answer: (a) $F_{AB} = mg\sqrt{2}$

- (b) Rod AB is under tension.